



CASO DE ESTUDIO

Proyecto INF3772

CONFIGURACIÓN SWITCHING



VLANs

El diseño de VLANs se realizó de la manera siguiente:

Vlan	ID Vlan	Network ID	prefijo /
Walumnos	10	172.16.0.0	/21
TelefonialP	15	172.16.8.0	/23
Wdocentes	20	172.16.10.0	/24
Docentes	25	172.16.11.0	/24
Wadmin	30	172.16.12.0	/24
Admin	35	172.16.13.0	/24
Mgm	40	172.16.14.0	/25
CCTV	45	172.16.14.64	/25
Servidores	50	172.16.14.128	/27
Lab1	55	172.16.14.160	/27
Lab2	60	172.16.14.192	/27
Lab3	65	172.16.14.224	/27
Lab4	70	172.16.15.0	/27
Lab5	75	172.16.15.32	/27

Para visualizar la tabla de direccionamiento IPv4 completa revisar el documento:
“Diseño IPv4 Caso-Proyecto INF3772.pdf”

VTP

Se aplico el protocolo VTP en su forma básica y funcional:

- MS-D11: es el servidor VTP, pertenece al dominio *inf3772.com* , con el password *inf3772*

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```
MS-D11
Physical Config CLI Attributes
IOS Command Line Interface
<CR>
MS-D11#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : inf3772.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.C75D.6C00
Configuration last modified by 0.0.0.0 at 10-1-26 16:23:53
Local updater ID is 172.16.0.2 on interface Vll0 (lowest numbered VLAN interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 19
Configuration Revision   : 1405
MD5 digest               : 0x82 0xFB 0x42 0x4A 0xBE 0x5E 0x11 0x50
                        : 0xA5 0x5D 0xBF 0xDF 0x48 0x48 0xBC 0x5D

MS-D11#
MS-D11#
```

```
MS-D11
Physical Config CLI Attributes
IOS Command Line Interface
MS-D11#show vlan brief

VLAN Name                Status    Ports
-----
1    default              active    Gig1/0/1, Gig1/0/2, Gig1/0/3, Gig1/0/4
                                Gig1/0/5, Gig1/0/6, Gig1/0/7, Gig1/0/8
                                Gig1/0/9, Gig1/0/11, Gig1/0/12, Gig1/0/13
                                Gig1/0/14, Gig1/0/18, Gig1/0/19, Gig1/0/24
                                Gig1/1/2, Gig1/1/3, Gig1/1/4
10   Walumnos             active
15   TelefoníaIP          active
20   Wdocentes            active
25   Docentes             active
30   Wadmin               active
35   Admin                active
40   Mgm                  active    Gig1/0/10
45   CCTV               active
50   Servidores           active
55   Lab1                 active
60   Lab2                 active
65   Lab3                 active
```

Puertos Acceso y troncales

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En los switches de acceso se configuro las interfaces como acceso a la VLAN correspondiente, la VLAN de VOZ se configuro en las interfaces que pertenecen a las Vlan de Administracion y Docentes.

Los enlaces up-links hacia los switches de distribución son troncales configurados manualmente y desactivando la negociación DTP. La VLAN nativa es la VLAN40 Mgm.

SA1-ADC

```
SA1-ADC#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	Walumnos	active	
15	TelefoniaIP	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14
20	Wdocentes	active	
25	Docentes	active	Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20
30	Wadmin	active	
35	Admin	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14
40	Mgm	active	
45	CCTV	active	Fa0/21, Fa0/22, Fa0/23

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```
SA1-ADC#sh
SA1-ADC#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    40
Gig0/1    on        802.1q         trunking    40
Gig0/2    on        802.1q         trunking    40

Port      Vlans allowed on trunk
Fa0/24    1-1005
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Fa0/24    1,10,15,20,25,30,35,40,45,50,55,60,65,70,75
Gig0/1    1,10,15,20,25,30,35,40,45,50,55,60,65,70,75
Gig0/2    1,10,15,20,25,30,35,40,45,50,55,60,65,70,75

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    1,10,15,20,25,30,35,40,45,50,55,60,65,70,75
Gig0/1    1,10,20,30,40,50,60,70,75
Gig0/2    15,25,35,40,45,55,65,70
```

El puerto FastEthernet 0/24 de cada switch de acceso es troncal y esta conectado un LAP que transmite las Wlan en su VLAN correspondiente.

```
SA1-ADC#show interfaces fastEthernet 0/24 switchport
Name: Fa0/24
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 40 (Mgm)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
```

Rapid PVST+

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Se aplica una instancia de RSTP por Vlan, con las siguientes consideraciones:

- **MS-D11:** Switch de capa 3 dispuesto en la capa de distribución, es el *root primary* en la topología de RSTP para las VLANs 10,20,30,40,60,70 y como *root secondary* para las VLANs 15,25,35,45,55,65.
- **MS-D22:** Switch de capa 3 dispuesto en la capa de distribución, es el *root primary* en la topología de RSTP para las VLANs 15,25,35,45,55,65 y como *root secondary* para las VLANs 10,20,30,40,60,70.

```
MS-D11#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     000C.8507.2A31
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    (priority 24576 sys-id-ext 10)
             Address     000C.8507.2A31
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Gi1/0/20 Desg FWD 4 128.20 P2p
Gi1/0/22 Desg FWD 4 128.22 P2p
Gi1/0/21 Desg FWD 4 128.21 P2p
Gi1/0/23 Desg FWD 4 128.23 P2p
Po1 Desg FWD 3 128.29 P2p

MS-D22#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     000C.8507.2A31
             Cost         3
             Port         29 (Port-channel)
             Hello Time 2 sec Max Age 20 sec Forward Delay 15
             Aging Time 20

  Bridge ID  Priority    (priority 28672 sys-id-ext 10)
             Address     0060.70B6.BD62
             Hello Time 2 sec Max Age 20 sec Forward Delay 15
             Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Gi1/0/21 Desg FWD 4 128.21 P2p
Gi1/0/20 Desg FWD 4 128.20 P2p
Gi1/0/23 Desg FWD 4 128.23 P2p
```

- **MS-C11:** Switch de capa 3 para la VLAN 50 “Servidores” como *root primary*.
- **MS-C22:** Switch de capa 3 para la VLAN 50 “Servidores” como *root secondary*.

```
MS-C11#show spanning-tree via
MS-C11#show spanning-tree vlan 50
VLAN0050
  Spanning tree enabled protocol rstp
  Root ID    Priority    24626
             Address     0040.0B48.C6EB
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    (priority 24576 sys-id-ext 50)
             Address     0040.0B48.C6EB
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Gi1/0/1 Desg FWD 4 128.1 P2p

MS-C22#show spanning-tree vlan 50
VLAN0050
  Spanning tree enabled protocol rstp
  Root ID    Priority    24626
             Address     0040.0B48.C6EB
             Cost         8
             Port         1 (GigabitEthernet1/0/1)
             Hello Time 2 sec Max Age 20 sec Forward Delay 15
             Aging Time 20

  Bridge ID  Priority    (priority 28672 sys-id-ext 50)
             Address     0001.9620.33BC
             Hello Time 2 sec Max Age 20 sec Forward Delay 15
             Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Gi1/0/1 Root FWD 4 128.1 P2p
```

Porfast y BPDUGuard en los puertos de los switches de acceso que se conectan a los dispositivos finales.

Port-channel

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- L2: Po1 entre los switches de distribución en los enlaces GigabitEthernet 1/0/15 – 17 es un enlace troncal, ambos extremos están configurados en modo activo formando el PO1 en LACP.

```
MS-D11#show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----
1      Po1(SU)        LACP        Gig1/0/15(P) Gig1/0/16(P) Gig1/0/17(P)
MS-D11#
```

```
MS-D22#show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----
1      Po1(SU)        LACP        Gig1/0/15(P) Gig1/0/16(P) Gig1/0/17(P)
MS-D22#
```

```
MS-D11#show intr
MS-D11#show inte
MS-D11#show interfaces tr
MS-D11#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Po1       on        802.1q         trunking    40
Gig1/0/20 on        802.1q         trunking    40
Gig1/0/21 on        802.1q         trunking    40
Gig1/0/22 on        802.1q         trunking    40
Gig1/0/23 on        802.1q         trunking    40

Port      Vlans allowed on trunk
Po1       1-1005
Gig1/0/20 1-1005
Gig1/0/21 1-1005
Gig1/0/22 1-1005
Gig1/0/23 1-1005

Port      Vlans allowed and active in management domain
Po1       1,10,15,20,25,30,35,40,45,50,55,60,65,70,75
Gig1/0/20 1,10,15,20,25,30,35,40,45,50,55,60,65,70,75
MS-D11#
```

- L3 : Po2 entre los switches de núcleo en los enlaces GigabitEthernet 1/0/20 – 22 es un enlace troncal configurado de manera estática ON, y se le asigno el segmento de red 172.16.15.72/30, forman parte del proceso OSPFv2 enrutando la VLAN50.

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CONFIGURACIÓN SWITCHING



```
MS-C11#show eth
MS-C11#show etherchannel sum
MS-C11#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone S - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

Number of channel-groups in use: 1
Number of aggregators: 1

Group Port-channel Protocol Ports
-----
2 Po2 (RU) - Gig1/0/20(P) Gig1/0/21(P) Gig1/0/22(P)
MS-C11#

MS-C22#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone S - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

Number of channel-groups in use: 1
Number of aggregators: 1

Group Port-channel Protocol Ports
-----
2 Po2 (RU) - Gig1/0/20(P) Gig1/0/21(P) Gig1/0/22(P)
MS-C22#

MS-C11#show ip interface brief
Interface IP-Address OK? Method Status Protocol
Port-channel2 172.16.15.73 YES manual up up
GigabitEthernet1/0/1 unassigned YES unset up up
GigabitEthernet1/0/2 unassigned YES unset down down
GigabitEthernet1/0/3 unassigned YES unset down down
GigabitEthernet1/0/4 unassigned YES unset down down
GigabitEthernet1/0/5 unassigned YES unset down down
GigabitEthernet1/0/6 unassigned YES unset down down
GigabitEthernet1/0/7 unassigned YES unset down down
GigabitEthernet1/0/8 unassigned YES unset down down
GigabitEthernet1/0/9 unassigned YES unset down down
GigabitEthernet1/0/10 unassigned YES unset down down
GigabitEthernet1/0/11 unassigned YES unset down down
GigabitEthernet1/0/12 unassigned YES unset down down
GigabitEthernet1/0/13 unassigned YES unset down down
MS-C11#

MS-C22#show ip int brief
Interface IP-Address OK? Method Status
Protocol
Port-channel2 172.16.15.74 YES manual up
up
GigabitEthernet1/0/1 unassigned YES NVRAM up
up
GigabitEthernet1/0/2 unassigned YES NVRAM down
down
GigabitEthernet1/0/3 unassigned YES NVRAM down
down
GigabitEthernet1/0/4 unassigned YES NVRAM down
down
GigabitEthernet1/0/5 unassigned YES NVRAM down
down
GigabitEthernet1/0/6 unassigned YES NVRAM down
down
GigabitEthernet1/0/7 unassigned YES NVRAM down
down
GigabitEthernet1/0/8 unassigned YES NVRAM down
down
GigabitEthernet1/0/9 unassigned YES NVRAM down
down
GigabitEthernet1/0/10 unassigned YES NVRAM down
down
GigabitEthernet1/0/11 unassigned YES NVRAM down
down
GigabitEthernet1/0/12 unassigned YES NVRAM down
down
GigabitEthernet1/0/13 unassigned YES NVRAM down
down
MS-C22#
```

HSRP

Como protocolo de redundancia en el primer salto se configuro HSRP una instancia para cada VLAN, en la capa de núcleo para la VLAN50 Servidores y en la capa de distribución para las demás VLANs.

```
Switch01#
!
interface Vlan50
 mac-address 0040.0b48.c602
 ip address 172.16.14.130 255.255.255.224
 ip ospf 1 area 0
 standby 50 ip 172.16.14.129
 standby 50 priority 110
 standby 50 preempt
!
router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 network 172.16.15.0 0.0.0.255 area 0
 network 172.16.14.0 0.0.0.255 area 0
 network 172.16.15.64 0.0.0.31 area 0
 network 172.16.14.128 0.0.0.31 area 0
 network 172.16.14.0 0.0.0.63 area 0
!
ip classless
!
ip flow-export version 9
!

Switch02#
!
interface Vlan50
 mac-address 0001.9620.3302
 ip address 172.16.14.131 255.255.255.224
 ip ospf 1 area 0
 standby 50 ip 172.16.14.129
!
router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
 network 172.16.14.0 0.0.0.255 area 0
 network 172.16.15.0 0.0.0.255 area 0
 network 172.16.15.68 0.0.0.31 area 0
 network 172.16.14.128 0.0.0.31 area 0
 network 172.16.14.0 0.0.0.63 area 0
!
ip classless
!
ip flow-export version 9
!

```

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```
MS-D11#show standby bri
MS-D11#show standby brief
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby Virtual IP
V110 10 100 Standby 172.16.0.3 local 172.16.0.1
V115 15 100 Active local 172.16.8.3 172.16.8.1
V120 20 100 Standby 172.16.10.3 local 172.16.10.1
V125 25 100 Standby 172.16.11.3 local 172.16.11.1
V130 30 100 Active local 172.16.12.3 172.16.12.1
V135 35 100 Standby 172.16.13.3 local 172.16.13.1
V140 40 100 Active local 172.16.14.3 172.16.14.1
V145 45 100 Active local 172.16.14.67 172.16.14.65
V155 55 100 Standby 172.16.14.163 local 172.16.14.161
V160 60 100 Active local 172.16.14.195 172.16.14.193
V165 65 100 Active local 172.16.14.227 172.16.14.225
V170 70 100 Standby 172.16.15.3 local 172.16.15.1
V175 75 100 Standby 172.16.15.35 local 172.16.15.33
MS-D11#

MS-D22#sh stan b
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby
Virtual IP
V110 10 100 Active local 172.16.0.2
172.16.0.1
V115 15 100 Standby 172.16.8.2 local
172.16.8.1
V120 20 100 Active local 172.16.10.2
172.16.10.1
V125 25 100 Active local 172.16.11.2
172.16.11.1
V130 30 100 Standby 172.16.12.2 local
172.16.12.1
V135 35 100 Active local 172.16.13.2
172.16.13.1
V140 40 100 Standby 172.16.14.2 local
172.16.14.1
V145 45 100 Standby 172.16.14.66 local
172.16.14.65
```

Para visualizar las configuraciones de los dispositivos diríjase a la carpeta configuraciones.

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